DAY 02 OF 10

Operators & Strings

Do maths, make decisions with comparisons and master text handling

Today's agenda

1. Arithmetic operators
2. Comparison and logical operators
3. Operator precedence
4. Strings: indexing and slicing
5. String methods and f-string formatting

Python in 10 Days
Lakshmi Prasad Gondinigari

Arithmetic operators

DAY 02

- + - * work as expected; / always gives a float
- // floor division, % remainder, ** power
- Use += -= *= as shortcuts: total += 5
- round(x, 2) rounds a number to 2 decimals

Python

```python
a, b = 17, 5
print(a + b)    # 22
print(a - b)    # 12
print(a * b)    # 85
print(a / b)    # 3.4
print(a // b)   # 3
print(a % b)    # 2
print(a ** b)   # 1419857
```

Comparison and logical operators

DAY 02

Operator	Meaning	Example	Result
== !=	equal, not equal	5 == 5	True
< > <= >=	greater / less than	3 >= 4	False
and	both must be True	True and False	False
or	at least one True	True or False	True
not	reverses the result	not True	False
in	membership test	"a" in "cat"	True

= assigns a value. == compares two values. Mixing them up is the most common beginner bug.

Using comparisons

- Comparisons produce True or False (bool values)
- Chain with and / or / not to build conditions
- Chained comparison is allowed: 18 <= age < 60
- These are used in if statements and loops (Day 3)

Python

```python
age = 20
print(age >= 18)              # True
print(age == 21)              # False
print(age > 18 and age < 60)  # True
print(not (age > 30))         # True
print(18 <= age < 60)         # True
```

Operator precedence

- Order of evaluation, highest first:
  1. Parentheses ( )
  2. Power **
  3. Multiply, divide, floor, modulo * / // %
  4. Add, subtract + -
  5. Comparisons, then not, and, or
- When in doubt, add brackets. They make code easier to read

Python

```python
print(2 + 3 * 4)       # 14
print((2 + 3) * 4)     # 20
print(2 ** 3 ** 2)     # 512  (right to left)
print(10 - 4 - 3)      # 3
print(True or False and False)  # True
```

Strings: the basics

- A string is a sequence of characters in quotes
- Index starts at 0. Negative index counts from the end
- Slicing: s[start:stop:step] (stop is not included)
- Strings are immutable: you cannot change one character in place
- len(s) gives the length

Python

```python
s = "Python"
print(s[0])      # P
print(s[-1])     # n
print(s[0:3])    # Pyt
print(s[2:])     # thon
print(s[::-1])   # nohtyP
print(len(s))    # 6
```

String methods (1)

- Methods are actions that belong to a string: s.method()
- They return a NEW string; the original is unchanged
- strip() removes spaces at both ends
- upper(), lower(), title(), capitalize() change case

Python

```
text = "  hello python world  "
print(text.strip())
print(text.strip().upper())
print(text.strip().title())
print(text.replace("python", "Python"))
```

OUTPUT

```
hello python world
HELLO PYTHON WORLD
Hello Python World
  hello Python world
```

String methods (2)

DAY 02

- split() breaks text into a list; join() does the reverse
- find() gives the position (or -1); count() counts occurrences
- startswith() / endswith() test the edges
- isdigit(), isalpha() check the content
- in checks if a piece of text exists

Python

```python
words = "red green blue".split()
print(words)
print("-".join(words))
print("banana".count("a"))
print("banana".find("nan"))
print("123".isdigit())
print("py" in "python")
```

OUTPUT

```
['red', 'green', 'blue']
red-green-blue
3
2
True
True
```

f-string formatting

- Put an expression inside { } within f"..."
- :.2f two decimal places
- :>10 right align in width 10; :<10 left; :^10 centre
- :, thousands separator

Python

```python
price = 49.5
qty = 3
print(f"Total: {price * qty}")
print(f"Price: {price:.2f}")
print(f"{'Python':>10}|")
print(f"{1234567:,}")
```

OUTPUT

```
Total: 148.5
Price: 49.50
    Python|
1,234,567
```

Escape sequences and multi-line text

- \n new line, \t tab, \\ backslash, \" quote
- Triple quotes """ """ allow text over many lines
- Raw string r"C:\new" keeps backslashes as they are
- Repeat text with *: "-" * 20

Python

```
print("Name\tMarks")
print("Asha\t82")
print("-" * 12)
print("""Line 1
Line 2""")
```

OUTPUT

```
Name Marks
Asha 82
------------
Line 1
Line 2
```

Practice time

DAY 02

1. Ask for two numbers; print +, -, *, /, // and % results
2. Check if a number is even using the % operator
3. Take a name and print it reversed, in uppercase and its length
4. Take a sentence and print the number of words
5. Print the first and last character of a word

Challenge: ask for a full name and print the initials ("lakshmi prasad" gives "L.P.")

Day 2 recap

DAY 02

- Arithmetic: + - * / // % **; comparisons give True / False
- Combine conditions with and / or / not
- Strings: indexing, slicing, immutable
- Methods: strip, upper, lower, split, join, replace, find, count
- Format output with f-strings

Homework: finish practice. Next: Conditions and Loops